3.1 Socio-technical Motivation

We systematically study of off-the-shelf instruction-tuned models across three persona categories—moral philosophies (e.g., utilitarianism), personality traits, and political ideologies (e.g., liberalism)— to determine if such constructs emerge as patterns in hidden-state activations, and (if so) precisely where they are located within the model's representational space.

The Big Five (openness, conscientiousness, extraversion, agreeableness, neuroticism) provide a well-validated framework for describing individual behavioral regularities or personality traits, which distinguish persons that are invariant over time and across situations [54, 55]. McCrae and Costa [56] argue that this framework shows the stability and consistency of personality traits, which help to predict how people will behave over time when placed in different situations. In human contexts, these traits correlate with patterns of decision-making [57, 58], social interaction [59], and even vulnerability to manipulation [27, 60, 61]. A growing body of work has leveraged the Big Five personality model to better understand LLM behavior, e.g., how prompts with personality trait information influences the output [62], the capacity of LLMs to infer Big Five personality traits from dialogues [63], the behavior of persona-instructed LLMs on personality tests, and creative writing tasks [38].

As LLMs are increasingly used to support decision-making in high-stakes scenarios, it is important to understand which ethical perspective governs a model's proposed solution. Prior work has examined the alignment between LLM-driven decisions and human moral judgments through the lens of persona-based prompting [64, 65], showing, for instance, that political personas can significantly influence model behavior in ethical dilemmas [65]. Other studies have developed benchmarks and systematic evaluations to assess the moral identity and decision-making patterns of LLMs [66, 67]. Others found fine-tuning to be a promising strategy for aligning LLM agents more closely with human values [68].

LLMs exposed to vast amounts of political text may internalize subtle political biases [9, 23]. Such systematic leanings, e.g., when LLMs are used in chatbots, summarizers, and recommendation systems, can undermine democratic discourse, skew the information ecosystem, and disenfranchise minority viewpoints [69]. Recent studies have shown that instruction-tuned LLMs can simulate divergent political viewpoints—sometimes classifying the same news outlet differently across runs—and often disagree with expert or human-annotated stances [70].

Finally, personality, ethics, and politics do not operate in isolation. For example, a model's moral reasoning may shift when presented through a particular political lens (e.g., justifications for decisions can be different for a conservative utilitarian than for a liberal utilitarian persona). Existing work has examined the output of LLMs with regard to personas with different combinations of demographics [39, 71, 72]. Here, as a first step toward exploring the intersections of personas across ethical, political, and personality dimensions, we examine the overlap in their embeddings.

3.2 Persona Datasets and Assumptions

Our experiments are based on the model-generated personas [3], consisting of statements written from the perspective of individuals with specific personalities, beliefs, or viewpoints (e.g., *extraversion* and *agreeableness*).[2] Each statement has an associated (model-generated) label indicating whether it matches the behavior of the corresponding persona dimension. For example, in the *extraversion* dataset, the sentence "Lively, adventurous, willing to take risks" is labeled as MATCHINGBEHAVIOR, whereas "I am quiet and don't socialize much" is labeled as NOTMATCHINGBEHAVIOR. As discussed in [3], an LLM was used to generate both the label and an associated confidence score. Detailed descriptions of the methodology used for the generation of these statements, the labeling process, and verification can be found in the original paper [3].

Persona Dimensions. Our work analyzes personas across three categories: personality, ethical theories, and political views. In this work, we examine three subsets of those topics, resulting in fourteen datasets:

- *Primary Personality Dimensions*, which relies on the Big Five [34, 54], a widely recognized framework for understanding human behavior and interpersonal dynamics. The five personas considered are characterized by *agreeableness* (AGREE), *conscientiousness* (CONSC), *openness* (OPEN), *extraversion* (EXTRA), and *neuroticism* (NEURO).
- *Ethical Theories*, which influence moral reasoning and value-based judgment and are central to decision-making and social interactions. The five personas considered are *subscribes-to-virtue-ethics* (VIRTUE), *subscribes-to-culturalrelativism* (RELAT), *subscribes-to-deontology* (DEONT), *subscribes-to-utilitarianism* (UTILI), and *subscribes-to-moralnihilism* (NIHIL).

[2]For a critical discussion on synthetic persona generation refer to Haxvig [73].

- *Political Views*, which capture ideological leanings and political affiliations, reflecting individuals' values and preferences for societal order. We consider a subset of four personas:[3] *politically-conservative* (CONS), *politically-liberal* (LIBER), *anti-immigration* (IMMI), and *anti-LGBTQ-rights* (LGBTQ).

For examples of each of the above personas and more explanations, refer to Appendix Tab. 3. By examining these three topics of personas, we can uncover the interplay between these facets.

Following [5], we filter the original dimensions based on a minimum number of sufficiently confident examples. We impose a minimum label confidence of 0.85 and a minimum count of 300 for each direction, MATCHINGBEHAVIOR and NOTMATCHINGBEHAVIOR, of each persona dimension, resulting in a total of 600 examples per persona dataset.

Assumptions of the Dataset. A key assumption underlying the dataset, which is crucial for interpreting our study, is that a NOTMATCHINGBEHAVIOR label does not inherently imply MATCHINGBEHAVIOR for a different persona.[4] For example, a statement from the *politically-liberal* dataset labeled as NOTMATCHINGBEHAVIOR does not automatically mean it aligns with sentences in the *politically-conservative* dataset. Consider the sentence: "Women should be subservient to men." This appears in the *politically-liberal* dataset with a NOTMATCHINGBEHAVIOR label, but this does not imply that a similar sentence exists in the *politically-conservative* dataset (and, in fact, does not). At the same time, we do observe some overlap between persona datasets. For instance, the sentence "I support marriage equality and LGBTQ rights." is labeled as MATCHINGBEHAVIOR in the *politically-liberal* dataset and NOTMATCHINGBEHAVIOR in the *anti-LGBTQ-rights* dataset. It is crucial to understand that the label does not indicate movement along a continuous axis but instead indicates the presence of a behavior.

3.3 Selection of LLMs and Embedding Vectors

We study the internal representations of three models, *Llama3-8B-Instruct* (*Llama3*) [74], *Granite-7B-Instruct* (*Granite*) [75], and *Mistral-7B-Instruct* (*Mistral*) [76]. We focus on instruct models because, unlike base LLMs that rely on a next-word prediction objective, instruct models are fine-tuned specifically for instruction following [22]. They are typically trained using supervised fine-tuning with question-answer pairs annotated by human experts and reinforcement learning with human feedback, allowing them to learn which responses are most useful or relevant to humans [77]. As a result, these models are likely better trained to adhere to persona behaviors. Additionally, instruct models tend to exhibit more predictable behavior than base models [22], making them more reliable for controlled experiments.

We extract the representation vectors at each layer from each model's forward pass when processing MATCHINGBEHAVIOR and NOTMATCHINGBEHAVIOR statements for a given dimension. We only keep the vector corresponding to the last token of each sentence at each layer as it contains relevant and summarized information of the whole sentence [78]. All models considered in this study are decoder-only models with 32 layers, and the activation vector from the last token has a shape of $(1, 4096)$. For a more detailed description of the specific models, see Appendix A.2.

3.4 Research Questions

In this study, we aim to answer two key questions: **(Q1)** Where in the model are persona representations encoded? **(Q2)** How do these representations vary across different personas?

For **(Q1)**, we investigate which layers in LLMs exhibit the strongest signal for encoding persona-specific information. This is important because knowing the layer-wise distribution of persona features can provide better insights into how complex behavioral and human characteristics are encoded in the model. Such insights could drive improvements in model interpretability and enable targeted interventions. Prior work has shown that transformer architectures tend to localize different types of linguistic and semantic information in distinct layers [79], yet the encoding of persona-specific characteristics remains under-explored.

For **(Q2)**, we seek to determine whether there are consistent, unique locations within a given LLM layer where distinct persona representations are encoded. Uncovering such patterns is crucial to understanding whether persona features are confined to specific subspaces within the model. This finding could facilitate more effective methods for controlling and customizing LLM outputs according to desired persona traits. Previous research in neural network interpretability has identified specialized neurons for various linguistic functions [2]. Similar structures regarding persona representations have not yet been studied.

[3] We exclude BELIEVES-IN-GUNRIGHTS and BELIEVES-ABORTION-SHOULD-BE-ILLEGAL.

[4] Prompts asked for statements the persona "would agree with, but others would disagree with," where *others* refers to any persona not aligned with the one under consideration [3].

4 Localization of Persona Representations

We provide an overview of the methods used to localize persona representations in LLMs. We first describe the methods used to identify and validate the layer in a given model where the embeddings of a specific persona differ most from those of others (§ 4.1), then present the approach for identifying the subset of activations within that layer that play a critical role in encoding a particular persona compared to other personas (§ 4.2).

4.1 Identifying Layers With Strongest Persona Representations

To investigate where persona representations are encoded **(Q1)**, we aim to identify the model layer at which the embeddings for a specific persona (MATCHINGBEHAVIOR) deviate most from those of other personas (NOTMATCHINGBEHAVIOR).[5]

For a given layer, let e^+ represent the set of embedding vectors corresponding to MATCHINGBEHAVIOR sentences, and e^- represent the set of embedding vectors corresponding to NOTMATCHINGBEHAVIOR sentences. Given the high-dimensional nature of these embeddings, we perform dimensionality reduction and compute their principal components (PCs) over the combined set of embeddings ($e^+ \cup e^-$). We denote the embeddings in the PC space as q^+ for MATCHINGBEHAVIOR and q^- for NOTMATCHINGBEHAVIOR.[6] We use several clustering metrics to quantify the differences between these two sets. Thereby, we treat each set, q^+ and q^-, as a cluster and compute the following distance metrics and scores.[7] We report results in § 5 over five independent runs, each using $q^+ = q^- = 100$ randomly sampled data points.

Calinski-Harabasz Score. The score is defined as the ratio of the sum of between-cluster dispersion (BCD) and within-cluster dispersion (WCD) [81]. BCD measures how well clusters are separated from each other. WCD measures the cluster compactness or cohesiveness.

Silhouette Score. The score is calculated using the mean intra-cluster distance and the mean nearest-cluster distance for each sample [82]. Values near 0 would indicate that representations from q^+ and q^- overlap, thus indicating non-sufficient capabilities to capture the given dimension.

Davies-Bouldin Score. The score is defined as the average similarity metric of each cluster with its most similar cluster, where similarity is the ratio of within-cluster distances to between-cluster distances [80, 83]. Thus, farther apart and less dispersed clusters will result in a better score.

Euclidean Distance. We measure the Euclidean distance between centroids C^+ and C^-; where $C^j = \frac{\sum_{p \in Conv(q^j)} p}{|Conv(q^j)|}$, with the convex hull $Conv(q^j)$ as the minimal convex set containing all points p in q^j.

4.2 Identifying a Layer's Activations With Strongest Persona Representations

For our second question **(Q2)**, we examine whether there are consistent activation patterns—distinct groups within sentence embedding vectors—that systematically encode different personas within a given layer. Inspired by previous work [51, 53], we adopt Deep Scan to analyze systematic shifts in neural network activation spaces. For additional related work, see § 2. We now present the method formally.

Let an LLM encode a statement X_m at a layer into an activation vector e_m. For instance, e_m could be the last token embedding vector at the last layer of a *Llama3* model that takes as input the statement X_m : "I believe strongly in family values and traditions", which is a MATCHINGBEHAVIOR for the CONS dimension. Each activation vector e_m consists of J activation units e_{mj}. The positions in this activation vector form the set of $O = \{O_1 \cdots O_J\}$ elements. Thus, J is the dimensionality of the embedding space, e.g., for *Llama3*, $J = 4096$ (see § 3.3). Consider a set of statements from a given persona dataset (e.g., CONSC), denoted as $X = \{X_1, \ldots, X_M\}$. Let $X_S \subseteq X$ and $O_S \subseteq O$, then we define a subset as $S = X_S \times O_S$. We call this a subset of sentences and activations. Our goal is to find the most persona-specific subset. To do this, we use a score function $F(S)$, which quantifies the anomalousness of a subset S. For instance, given the CONS dataset, the scoring function $F(S')$ with $S' = \{X_m\} \times \{O_j\}$ measures how divergent the last token representation of a sentence X_m, is at a given embedding position O_j, compared to the last token representations of all other sentences that are labeled MATCHINGBEHAVIOR. Thus, Deep Scan seeks to find the most salient subset of

[5] See the assumptions of the dataset in § 3.2.

[6] Explained variance ratio across 14 dimensions is 0.657 to 0.898.

[7] We use the scikit-learn implementations [80].